

Unit-II Dictionaries

Implementation of dictionary methods

1) dict.clear():-

The method **clear()** removes all items from the dictionary.

Syntax

```
dict.clear()
```

Return Value

This method does not return any value.

Example

```
dict = {'Name': 'Zara', 'Age': 7};  
dict.clear()
```

2) dict.copy():-

The method **copy()** returns a copy of the dictionary.

Syntax

```
dict.copy()
```

Example

```
dict1 = {'Name': 'Zara', 'Age': 7};  
dict2 = dict1.copy()  
print(dict2)
```

Output:-

```
{'Age': 7, 'Name': 'Zara'}
```

3) dict.fromkeys():-

The method **fromkeys()** creates a new dictionary with keys from *seq* and *values* set to value.

Syntax

```
dict.fromkeys(seq[, value])
```

Parameters

- **seq** – This is the list of values which would be used for dictionary keys preparation.
- **value** – This is optional, if provided then value would be set to this value

Return Value

This method returns the list.

Example

```
seq = ('name', 'age', 'sex')  
dict = dict.fromkeys(seq)  
print (dict)
```



```
dict = dict.fromkeys(seq, 10)
print (dict)
```

Output:-

```
{'age': None, 'name': None, 'sex': None}
{'age': 10, 'name': 10, 'sex': 10}
```

4) **dict.get(key, default=None):-**

The method **get()** returns a value for the given key. If key is not available then returns default value None.

Syntax

```
dict.get(key, default = None)
```

Parameters

- **key** – This is the Key to be searched in the dictionary.
- **default** – This is the Value to be returned in case key does not exist.

Return Value

This method return a value for the given key. If key is not available, then returns default value None.

Example

```
dict = {'Name': 'Zabra', 'Age': 7}
print dict.get('Age')
print dict.get('Education', "Never")
```

Output:-

```
7
```

```
Never
```

5) **dict.items():-**

The method **items()** returns a list of dict's (key, value) tuple pairs

Syntax

```
dict.items()
```

Return Value

This method returns a list of tuple pairs.

Example

```
dict = {'Name': 'Zara', 'Age': 7}
print dict.items()
```

Output:-

```
[('Age', 7), ('Name', 'Zara')]
```

6) **dict.keys():-**

The method **keys()** returns a list of all the available keys in the dictionary.

Syntax

```
dict.keys()
```



Return Value

This method returns a list of all the available keys in the dictionary.

Example

```
dict = {'Name': 'Zara', 'Age': 7}
print dict.keys()
```

Output:-

```
['Age', 'Name']
```

7) dict.update(dict2):-

The method **update()** adds dictionary dict2's key-values pairs in to dict. This function does not return anything.

Syntax

```
dict.update(dict2)
```

Parameters

- **dict2** – This is the dictionary to be added into dict.

Return Value

This method does not return any value.

Example

```
dict = {'Name': 'Zara', 'Age': 7}
dict2 = {'Sex': 'female'}
dict.update(dict2)
print (dict)
```

Output:-

```
{'Age': 7, 'Name': 'Zara', 'Sex': 'female'}
```

8) dict.values():-

The method values() returns a list of all the values available in a given dictionary.

Syntax

```
dict.values()
```

Return Value

This method returns a list of all the values available in a given dictionary.

Example

```
dict = {'Name': 'Zara', 'Age': 7}
print (dict.values())
```

Output:-

```
[7, 'Zara']
```

9) dict.pop(key):-

Removes the element with the specified key

Syntax



```
dict.pop(key)
```

Example

```
dict = {'Name': 'Zara', 'Age': 7}  
x=dict.pop('Age')  
print (x)
```

10) **dict.popitem(key):-**

Removes the last inserted key-value pair

Syntax

```
dict.popitem()
```

Example

```
dict = {'Name': 'Zara', 'Age': 7}  
x=dict.popitem()  
print (x)
```

11) **dict.setdefault(key,default-val):-**

Returns the value of the specified key. If the key does not exist: insert the key, with the specified value

Key- key to be searched in dictionary

Default-val- key with a value i.e. default value inserted in dictionary if the key is not in the directory. If the default value is not provided it takes none value

Syntax

```
dict.setdefault(key,default-val)
```

Example

```
dict = {'Name': 'Zara', 'Age': 7}  
x=dict.setdefault('Age')  
print(x)  
x=dict.setdefault('Address')  
print (x)  
x=dict.setdefault('Address','Sangola')  
print (x)
```

